ENABLING UNIFIED CLINICAL EDUCATION OPERATIONS THROUGH AN INTEGRATED DIGITAL PLATFORM

CHALLENGE OWNER

The National University Hospital (NUH) is a leading healthcare institution in Singapore and the largest medical center within the National University Health System (NUHS). Established in 1985, NUH serves over one million patients annually with its 1,200 beds and offers a comprehensive range of medical, surgical, and dental specialties. As an academic medical centre, it functions as the principal teaching hospital for the NUS Yong Loo Lin School of Medicine, actively engaging in research and international collaborations. NUH is providing integrated care for adults, women, and children, including specialised services like the country's sole pediatric kidney and liver transplant program. With state-of-the-art facilities and a dedicated team of healthcare professionals, NUH continues to lead in patient care, medical education, and research, playing a crucial role in advancing Singapore's healthcare landscape.

CONTEXT

NUH delivers clinical education across Medicine, Allied Health, Nursing and other training programmes. While each discipline has distinct operational nuances, they share common core activities which are student onboarding and postings management, tutor engagement, teaching activity tracking, feedback collection and reporting. The Medical Affairs (Education) department at NUH plays a central coordinating role in delivering structured clinical education for medical, nursing and allied health students across multiple partner institutions such as SIT, NUS, Singapore Polytechnic and PSB Academy. Beyond facilitating student postings, the department manages teaching schedules, attendance tracking, assessment and feedback records, ensuring that data flows into downstream processes such as faculty appraisal, teaching awards, tutor billing and continuous programme quality improvement. As clinical education expands in scale and complexity, the department's operational responsibilities have grown significantly, requiring tighter coordination across disciplines, departments and institutions.

During clinical postings, students interact with multiple tutors across specialties, each contributing supervision hours and providing formative and summative feedback. Administrative teams must ensure that teaching activities and feedback are accurately captured, matched to the correct student-tutor pairings, and consolidated in a timely manner. This data supports critical downstream processes including student feedback summaries, faculty appraisal, teaching awards, tutor billing, accreditation reporting and continuous quality improvement initiatives. However, data management practices differ significantly across disciplines.

For Allied Health, Nursing and Training programmes, there is currently no dedicated digital management system. Administrative teams manage the full student lifecycle, from onboarding to offboarding, largely through manual Excel trackers and FormSG submissions. Midpoint feedback is often conducted through face-to-face check-ins, while end-of-posting evaluations are collected separately. Although workflows are broadly similar across programmes, small operational differences require additional manual handling. This fragmented approach has resulted in duplicated effort, inconsistent data standards, missed submissions, incorrect student-tutor matching, reporting delays and increased risk of errors.

For the Medical discipline, feedback and teaching data are collected through more structured systems. However, these systems remain fragmented and lack interoperability and flexibility. Managing large student cohorts, multiple rotations and frequent academic year transitions still requires substantial

manual coordination. Data inconsistencies across postings and departments remain a persistent challenge, limited transparency and efficiency.

Across all disciplines, the manual and fragmented processes create several challenges for NUH:

- Fragmented student lifecycle management. There is no integrated system to manage end-to-end pre-boarding, onboarding and offboarding processes, resulting in duplicated effort or higher risk of administrative errors.
- Disconnected and manual data management. Teaching and feedback data are stored across multiple unlinked platforms, requiring manual consolidation and reducing data accuracy and reliability.
- Inaccurate tracking of teaching hours. Teaching hours are manually logged or optionally submitted by students, leading to low compliance and inaccurate records. Event-by-event tagging of tutor and student interactions is also time-consuming.
- Inefficient feedback monitoring. Feedback collection lacks automation and real-time visibility, resulting in missed submissions, compliance gaps and delayed follow-ups. Administrators must also manually track feedback completion and send reminders to students.
- Reporting and analytics bottlenecks. Manual report generation and limited customisation capabilities create heavy administrative workload and constrain timely, data-driven decision-making.
- Survey overload and fatigue. During clinical postings, students interact with multiple tutors across specialties. Surveys currently are being sent for every tutor and student interaction regardless of how many sessions the student goes through in a single day. This results in survey fatigue and low participation rate amongst students.

Several solutions have been trialled by NUH, including QR code-based surveys for collecting teaching hours and feedback, Microsoft Power Apps for structured data capture and the upgrade of internal system to the BLUE system (a shared system extended by NUS to NUH). However, these approaches did not resolve the core challenges as they were piecemeal enhancements that did not sufficiently streamline and integrate workflows, standardise data or provide a single, reliable source of truth across disciplines.

PROBLEM STATEMENT

How might we streamline and standardise fragmented clinical education workflows through a single, unified system that improves operational efficiency and reduces manual effort across disciplines?

WHAT ARE WE LOOKING FOR?

Manual consolidation remains common across the various NUH disciplines, increasing administrative workload and slowing decision-making for appraisal, tutor billing, awards and programme quality improvement. As clinical education volume grow and stakeholder expectations increase, there is a pressing need to move towards a consolidated, scalable system that can harmonise workflows, standardise data collection and provide a single source of truth across Medicine, Allied Health, Nursing and Training programmes.

The solution should meet the following criteria:

- Student and tutor lifecycle management. Centralise the management of student and tutor records across Medicine, Allied Health, Nursing and Training programmes. The solution should have the ability to assign student to departments, posting, tutors and academic cycles. There should be clear status tracking for onboarding, active and completed postings.
- Tracking of teaching hours. Provide a standardised and validated mechanism for capturing and verifying teaching contributions with the ability to tag sessions to specific students, posting

and departments. The solution should be able to allow tagging of approved hours to billable rates for compensation processing.

- Configurable survey templates. Allow configurable survey templates by discipline, posting type or semester, with the scoring matrix based out of 5. The solution should also have the ability to include compulsory text fields to capture reasons for surveys that receive low scores and flag them to the relevant users.
- Structured feedback collection. Standardise and streamline qualitative and quantitative feedback processes including real-time visibility of submission rates, automated reminders for incomplete submissions, tools to filter or categorise positive and negative feedback and the ability to generate customisable reports for supervisors, tutors, institutions for awards or appraisal purposes.
- Notifications for pending and incomplete tasks. Trigger alerts or notifications to the relevant users for tasks that are pending completion or approvals.
- Batch upload of student and tutor data. Support the batch upload of student and tutor data (i.e. from FormSG and Microsoft Excel) in each academic cycle with the ability to detect duplicated entries, errors and anomalies.
- Role-based access control. Must provide robust governance controls, with NUH as the primary owner. This includes role-based access controls (e.g. super admin, programme admin, supervisor, tutor), customisable permissions and approval hierarchies for handling the different approval workflows and viewership rights.
- Real-time dashboard and reporting capabilities. Provide a real-time dashboard of teaching hours by tutor, department and discipline as well as survey completion rates and deadlines. The solution should be capable of generating reports to support faculty appraisal, billing, awards and programme quality improvement.
- Maintenance of audit trail. Full audit trail of submissions, edits and approvals should be captured with timestamps for audit and compliance purposes.
- Data exportability function. Ability for administrators to download and export reports for finance, appraisal and accreditation purposes.
- Interface with existing information system. Ability to integrate with the existing hospital systems (e.g. NUHS Jobplan Portal/ NUS BLUE/ Teaching Activities System (TAS)) to ensure data completeness and seamless data flow, as part of the eventual deployment beyond the prototype/POC.

OVERALL PERFORMANCE REQUIREMENTS

- Accurate and efficient. Demonstrate accuracy in tracking teaching hours, reduce manual transcription and report generation time.
- User-friendly. Intuitive interface for clinicians, students and administrators.
- Configurable without heavy custom coding. Ability to adapt to different disciplines and future workflow changes without the need for heavy custom coding.
- Scalable. The solution should be scalable to handle large cohorts, growing student numbers, additional programmes and multi-institution data.
- Security and data privacy. Ensure that all staff login are secured and with access logs. All data should also be secured and comply with local data protection regulations (e.g. PDPA).
- Hosting on local cloud. The solution must be deployed in a Singapore-based cloud environment to meet local regulations and data residency requirements.

There are no restrictions on the geographical location of the problem solvers who may choose to apply to this challenge. However, the prototype must be demonstrated in Singapore.

METRICS OF SUCCESS

- Improved operation efficiency. Achieve a 50% reduction in time spent on manual data entry and tracking of teaching hours.
- Improved data accuracy. Achieve at least 90% data accuracy and completeness by consolidating and standardising data across all disciplines.
- Faster report generation rates. Reduction in report generation time by at least 80%, from the current five minutes to below one minute for appraisal, billing, awards and programme quality improvement purposes.
- Improved staff and user satisfaction rates. With the implementation of the solution, internal staff (administrators, tutors, clinicians) and user (students) satisfaction should increase, with at least a 30% increase in survey completion rates.

POSSIBLE USE CASES

1. Improved teaching hours and feedback consolidation. Sarah, who is with the Allied Health Education team, coordinates clinical postings for physiotherapy, occupational therapy and speech therapy students across multiple institutions. She manages onboarding, tracks teaching hours, collects feedback, prepares tutor evaluation reports and processes billing data. She maintains multiple Excel spreadsheets to track students, tutors and teaching hours. Feedback is collected via FormSG, which she manually downloads, filters and reformats into reports. She toggles between CSV files, emails and PDFs to reconcile records. Teaching hours are sometimes submitted late or inconsistently, and she has no easy way to verify accuracy. During peak periods, she spends days consolidating reports for tutor appraisals and awards.

   With an integrated platform, Sarah can view all student postings, tutor assignments, teaching hours and feedback submissions via a single dashboard. Teaching hours are logged directly by tutors and routed for approval, with automated anomaly flags. Feedback data is also automatically aggregated and report-ready. Instead of manual reconciliation, Sarah generates customised reports with one click, significantly reducing administrative workload and improving accuracy.

2. Fuss-free logging of teaching hours for medical tutors. Dr Aisha supervises medical students across different rotations, conducts formal teaching sessions and provides ad hoc consultations. She is required to log teaching hours and review student feedback. Teaching hours are currently logged through a cumbersome system, and additional hours must sometimes be submitted via email. Students optionally input their hours, leading to discrepancies. Dr Aisha has limited visibility of whether her hours have been approved or accurately recorded. Reviewing feedback requires navigating multiple screens and there is no consolidated view of her teaching contributions.

   Through a user-friendly interface, Dr Aisha logs both scheduled and ad hoc sessions within minutes. She can see her accumulated teaching hours in real time and track approval status. Feedback summaries are automatically compiled into a dashboard view, allowing her to quickly identify strengths and areas for improvement without administrative friction.

3. Better quality monitoring and strategic oversight. Bryan, the head of Medical Affairs, is responsible for overseeing clinical education across Medicine, Allied Health, Nursing and Training programmes. He oversees job shadowing applications, shortlists candidates, assigns clinician mentors and ensures teaching quality standards are met. He also tracks positive feedback for tutor recognition and identifies improvement opportunities from negative feedback. Student applications are reviewed manually and mentor-student matching is tracked in spreadsheets.

Positive feedback must be reformatted for award nominations, while negative feedback is manually extracted for programme review discussions. Data across discipline is fragmented, making it difficult to obtain a consolidated view of teaching contributions, feedback trends and compliance rates. Strategic decisions rely heavily on manually compiled reports that may already be outdated.

The solution provides Bryan with access to cross-disciplinary dashboards and standardised reporting. He gains real-time visibility into teaching hours, feedback completion rates and quality indicators. With reliable data at his fingertips, he can make informed decisions, strengthen governance and better support long-term programme improvement.

WHAT'S IN IT FOR YOU

- SGD50,000 of prize money to develop a Working Prototype, with the scope to be discussed and agreed upon between Challenge Owner and selected Solver, for each winner of this challenge (see Award Model)
- Access to IMDA's PIXEL corporate innovation hub and complimentary innovation consultancies (e.g. Design Thinking, Digital Storytelling) for the prototype development, where applicable
- Potential opportunities to implement the successfully developed and trialled prototype within NUH and possibly in the other NUH-affiliated medical institutions in Singapore

EVALUATION CRITERIA

The evaluation process shall take place over two stages. Proposals shall be evaluated based on the evaluation criteria set out for the first stage. Thereafter, shortlisted proposals shall be subjected to a second stage evaluation in the form of an interview / pitch, and the scoring shall be based on a re-defined assessment criteria for the selection of the challenge finalist(s).

Solution Fit (30%)	Relevance (30%): To what extent does the proposed solution address the problem statement effectively?
Solution Readiness (30%)	Maturity (15%): How ready is the proposed solution to go to the market? Scalability (15%): Is there any evidence to suggest capacity to scale?
Solution Advantage (20%)	Quality of Innovation (20%): Is the solution cost effective and truly innovative? Does it make use of new technologies in the market, and can it potentially generate new IP?
Company Profile (20%)	Business Traction (10%): Does the company have existing customers and revenue traction? Team Experience (10%): Do the team members possess strong scientific/technical background?

AWARD MODEL

30% of the prize money will be awarded to each selected finalist at the start of the POC/prototype development process. The remaining 70% will be awarded after completion of the POC/prototype solution, based on milestones agreed between Problem Owner(s) and the solver. Prize money will be inclusive of any applicable taxes and duties that any of the parties may incur.

*Note that a finalist who is selected to undertake the prototype will be required to enter into an agreement with Problem Owner(s) that will include more detailed conditions pertaining to the POC/prototype.

SUBMISSION GUIDELINES AND DEADLINE

All submissions should include sufficient information for assessment against the stated evaluation criteria. Problem Solvers are encouraged to include:

- **Solution proposal** addressing the problem statement (and any demos, where applicable)
- **Company business profile**, including past projects or relevant track record
- **Team profile** - information on the core team members involved in the project, their roles, technical certifications and relevant experience
- **Prototype development timeline** - outlining key milestones for prototype development and deployment
- **Scalability considerations**, including business model or estimated costs for future deployment, ongoing support and maintenance

All submissions must be made by **17 April 2026, 1600 hours (SGT/GMT +8)**. Problem Owner(s) and IMDA may extend the deadline of the submission at their discretion. Late submissions on the OIP, or submissions via GeBIZ, will not be considered.